1 A student investigates a spring of width w made from a metal wire, as shown in Fig. 1.1.

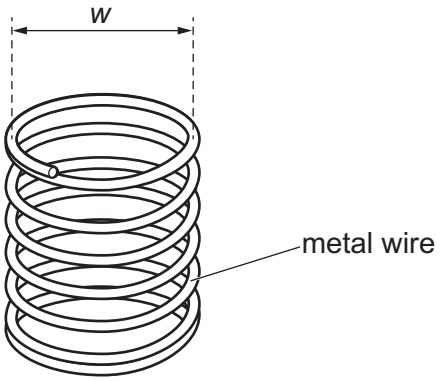

Fig. 1.1

The student constructs several springs, each made from a metal wire of different cross-sectional area A. The student investigates how the extension x of each spring varies with A when a load of mass m is applied.

It is suggested that the relationship between x and A is

$$x = \frac{mgw^3NA^n}{\gamma\rho}$$

where g is the acceleration of free fall, ρ is the density of the metal, N is the number of turns of wire in the spring and γ and n are constants.

Design a laboratory experiment to test the relationship between x and A.
Explain how your results could be used to determine values for γ and n.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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